Biochemistry Cume

Total points 100 points

1. A. Define K_M (**5 points**)
B. Define turnover number (**5 points**)
C. Write the Briggs-Haldane equation that depicts the kinetics of enzyme-catalyzed reactions.(**5 points**)
D. Based on the Briggs – Haldane equation, show that, (**10 points**)

$$K_M = \frac{k_{-1} +}{k_{1+}}$$

E. Calculate the catalytic time of one microgram of a pure enzyme (Molecular weight = 10 kDa) that catalyzes its substrate at a V_{max} of 0.1 mmoles/min. **(20 points)**

2. A. Define Exopeptidases **(5 points)**
B. Predict the products of treatment of the following polypeptide chain with chymotrypsin.

F-Q-Y-P-T-R-W-N-F-P-Y **(10 points)**

C. A polypeptide, OZARK, was subjected to the following enzymes and chemical reagent treatment. The products obtained after each such treatment is provided below. Deduce the amino acid sequence of the polypeptide, OZARK, based on the information provided below. [***Note – to receive credit complete alignment of all the products, that was used to deduce the amino acid sequence of the polypeptide chain, should be shown***] – (**15 points**)

a. Trypsin treatment of the polypeptide released the following products – **S-R , I-M-G-F-T-V, W-R, and. Y-M-S-T-Q-F-K.**
b. Chymotrypsin treatment of the polypeptide released the following products – **T-V, M-S-T-Q-F, W-R-S-R-Y, and K-I-M-G-F.**
c. Cyanogen bromide treatment of the polypeptide released the following products, **S-T-Q-F-K-I-M , G-F-T-V, and, W-R-S-R-Y-M.**

3. A. List the characteristic features of a sequential ordered reactions (**5 points**)
B. Provide an example of a double displacement/ping-pong reaction [***Note – to receive credit the name of the enzyme and the names and chemical structures of the reactants and products should be shown***] **(15 points**)
C. Depict the Cleland notation of a sequential random bi-substrate biochemical reaction (**10 points**)